Hands-On-Activity No. 2.2

Hands-on Activity 2.2: Control Structures (part 1)

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming logic and Design	**Date Performed: July 23, 2026**
Section: CPE11S2	**Date Submitted: July 24, 2026**
Name: Jose. Redge Jansen A.	**Instructor: Sir Noel Jason Lusung**

1. Objective(s)

This activity aims to develop programming skills using selection and repetition structures.

2. Intended Learning Outcomes

2.1 Understand and use conditions and conditional executions with the if/*else* selection structure

2.2 Understand and use conditions and conditional executions with the *while* selection structure

2.3 Understand and use counter controlled repetition and sentinel controlled repetition.

3. Materials and Equipment

Internet connection
Personal Computer

4. Procedure

C++ provides three types of selection structures.

- The **IF** selection structure either performs (selects) an action if a condition is true or skips the action if the condition is false.
- The **IF/ELSE** selection structure performs an action if a condition is true and performs a different action if the condition is false.
- The **SWITCH** selection structure performs one or many different actions depending on the value of an expression.

The **if** structure is called a *single-selection structure* because it selects or ignores a single action. The **if/else** structure is called a *double-selection structure* because it selects between two different actions. The **switch** structure is called a *multiple-selection structure* because it selects among many different actions.

Example of IF selection structure:

```
if (grade >= 50)

cout<<"Passed\n";
```

Example of IF/ELSE selection structure:

```
if (grade >= 50)

cout<<"Passed\n";

else
        cout<<"Failed\n";
```

Sample program: (cascaded statements)

```
#include<iostream>
using namespace std;
int main()
{
int grade;
cout<<"Enter grade:\n";
cin>>grade;
if(grade >= 90)
        cout<<"A\n";
else if (grade >= 80)
        cout<<"B\n";
else if (grade >= 70)
        cout<<"C\n";
else if (grade >= 60)
        cout<<"D\n";
else
        cout<<"F\n";
return 0;
}
```

The **while** repetition structure allows the programmer to specify that an action is to be repeated while some condition remains true.

Example pseudocode:

While there are more items on my shopping list

Purchase next item and cross it off my list

Example program:

```
#include<iostream>

using namespace std;

int main(void)

{

int product = 2;

while (product <= 1000)

        product = 2 * product;

        cout<<product;

}
```

5. Output

Exercise: Counter- Controlled Repetition. A class of ten students took a quiz. The grades (integers in the range of 0 to 100) for this quiz are available to you. Determine the class average on the quiz.

Set total to zero

Set grade counter to one

While grade counter is less than or equal to ten

Input the next grade

Add the grade into the total

Add one to the grade counter

Set the class average to the total divided by ten

Print the class average

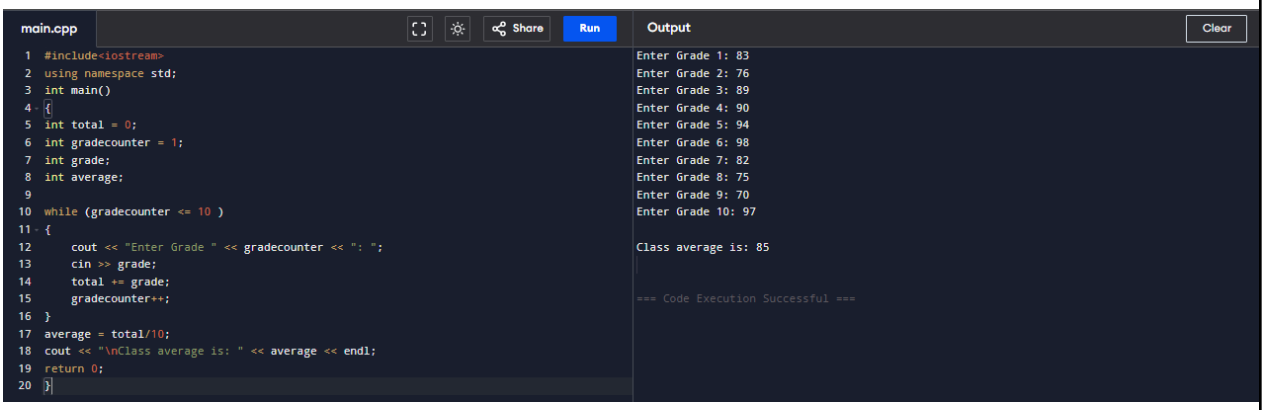
Counter controlled repetition is often called *definite repetition* because the number of repetitions is known before the loop begins executing. In the example above, the number of grades (10) was known in advance. What if there is no indication is given of how many grades are to be entered? How can the program determine when to stop the input of grades? How it will know when to calculate and print the class average?

One way to solve this is to use a special value called the sentinel value (also called a *signal value, a dummy xvalue or a flag value)* to indicate "end of data entry". The user types grades in until all legitimate grades have been entered. The user types the sentinel value to indicate that the last grade has been entered. Sentinel-controlled repetition is often called *indefinite repetition* because the number of repetitions is not known before the loop begins executing.

Answer:

```
#include<iostream>
using namespace std;
int main()
        {
        int total = 0;
        int gradecounter = 1;
        int grade;
        int average;

        while (gradecounter <= 10 )
        {
           cout << "Enter Grade " << gradecounter << ": ";
           cin >> grade;
           total += grade;
           gradecounter++;
        }
average = total/10;
cout << "\nClass average is: " << average << endl;
return 0;
}
```



Activity:

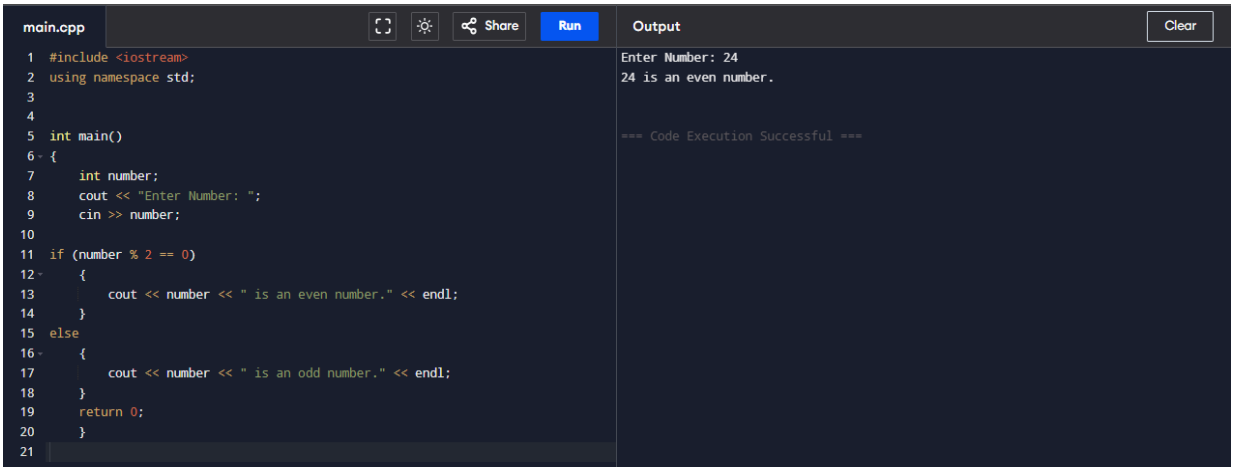
1. **Using conditional statements (if-else statements), write a program that asks a user for a number and prints out if it is an even or an odd number.**

```
#include <iostream>
using namespace std;

int main()
{
   int number;
   cout << "Enter Number: ";
   cin >> number;

if (number % 2 == 0)
   {
      cout << number << " is an even number." << endl;
```

```
    }
else
    {
        cout << number << " is an odd number." << endl;
    }
    return 0;
    }
```



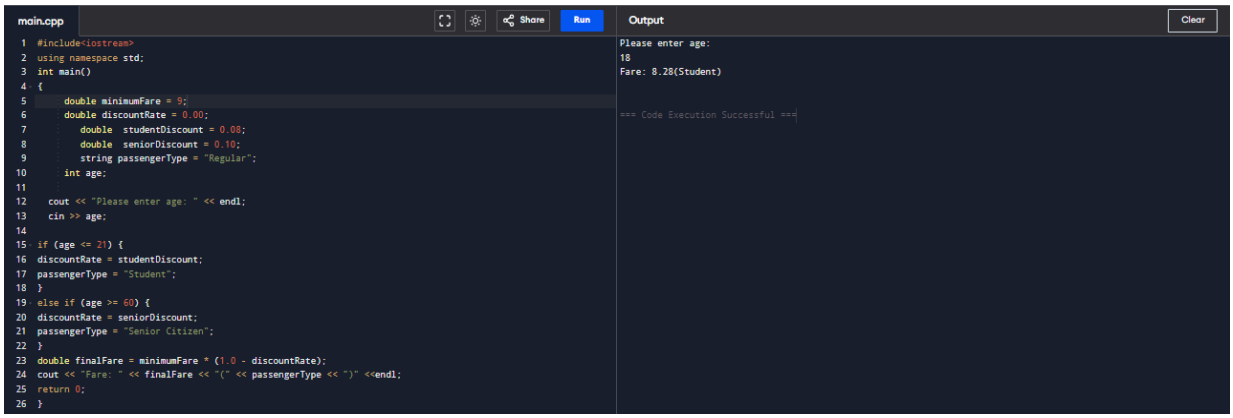
2. Using conditional statements, write a program that computes for 10 percent fare discount of a senior citizen and 8 percent fare discount of a student. There will be no discount if not a senior citizen and not a student. The user will be asked to enter age. The minimum fare is 9 pesos.

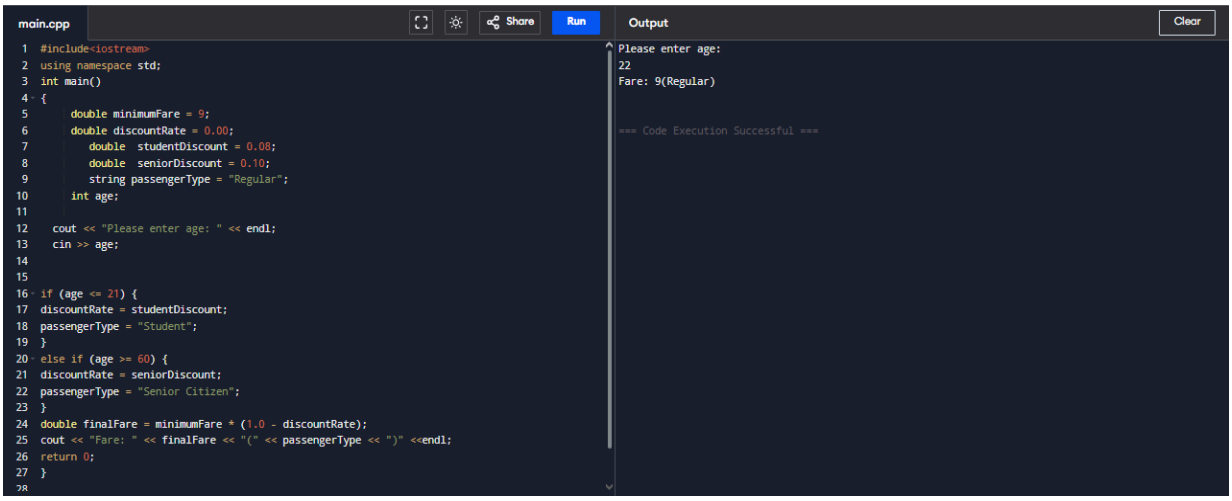
```
#include<iostream>
    using namespace std;
    int main()
    {
        double minimumFare = 9;
        double discountRate = 0.00;
          double  studentDiscount = 0.08;
          double  seniorDiscount = 0.10;
          string passengerType = "Regular";
        int age;

     cout << "Please enter age: " << endl;
     cin >> age;

    if (age <= 21) {
```

```
        discountRate = studentDiscount;
        passengerType = "Student";
        }
        else if (age >= 60) {
        discountRate = seniorDiscount;
        passengerType = "Senior Citizen";
        }
        double finalFare = minimumFare * (1.0 - discountRate);
        cout << "Fare: " << finalFare << "(" << passengerType << ")" <<endl;
return 0;
}
```





main.cpp

```cpp
#include<iostream>
using namespace std;
int main()
{
     double minimumFare = 9;
     double discountRate = 0.00;
        double  studentDiscount = 0.08;
        double  seniorDiscount = 0.10;
        string passengerType = "Regular";
     int age;

   cout << "Please enter age: " << endl;
   cin >> age;


if (age <= 21) {
discountRate = studentDiscount;
passengerType = "Student";
}
else if (age >= 60) {
discountRate = seniorDiscount;
passengerType = "Senior Citizen";
}
double finalFare = minimumFare * (1.0 - discountRate);
cout << "Fare: " << finalFare << "(" << passengerType << ")" <<endl;
return 0;
}
```

Output

```
Please enter age:
61
Fare: 8.1(Senior Citizen)


=== Code Execution Successful ===
```

3. **Case Study: Sentinel Controlled Repetition. Given the following pseudocode, create a program that will implement a sentinel controlled repetition. For example, you can use (-1) as the sentinel value. You can use Problem 1 as your reference.**

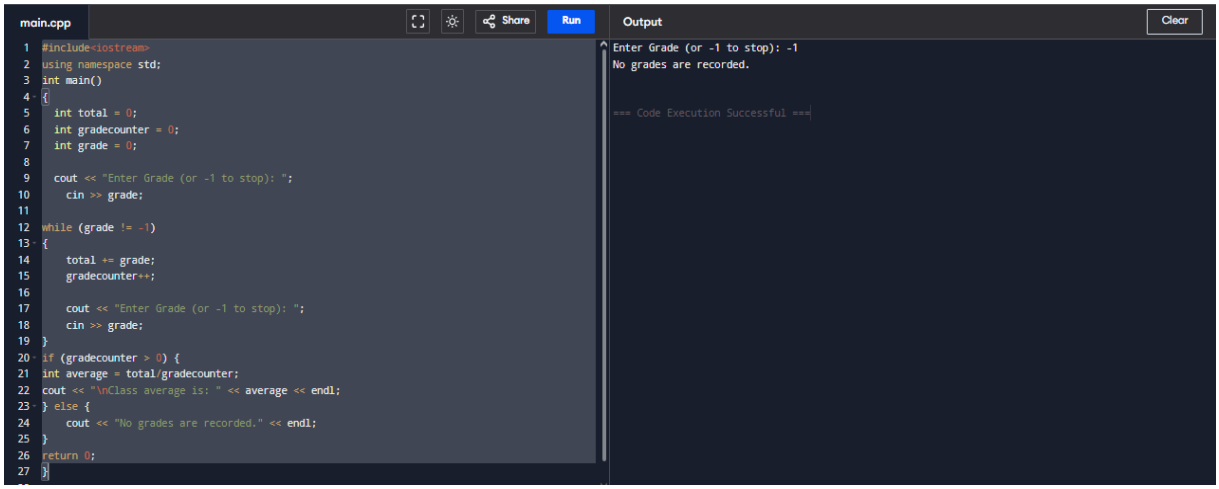
```
#include<iostream>
using namespace std;
int main()
{
  int total = 0;
  int gradecounter = 0;
  int grade = 0;

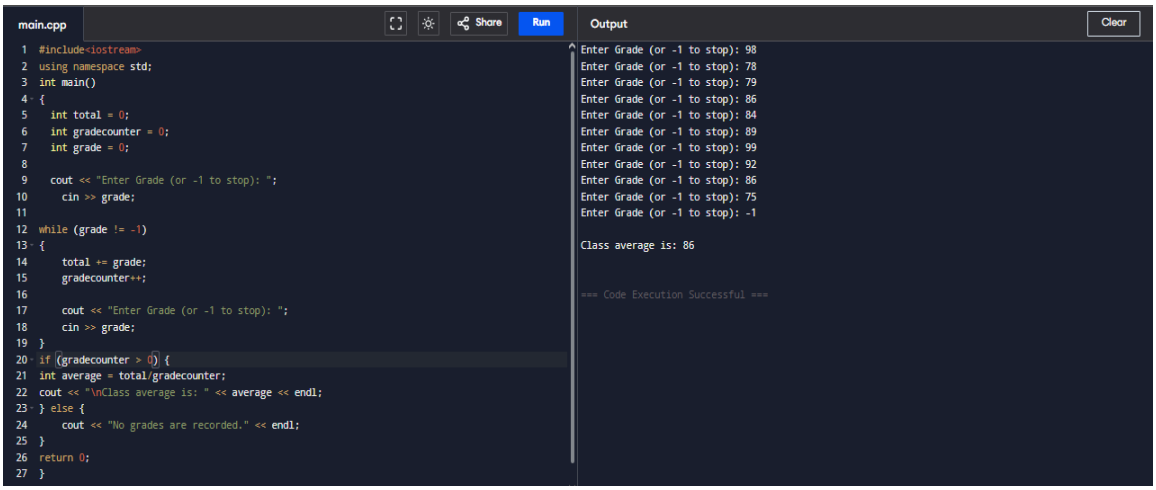
  cout << "Enter Grade (or -1 to stop): ";
    cin >> grade;

while (grade != -1)
{
    total += grade;
    gradecounter++;

    cout << "Enter Grade (or -1 to stop): ";
    cin >> grade;
}
if (gradecounter > 0) {
int average = total/gradecounter;
cout << "\nClass average is: " << average << endl;
} else {
```

```
        cout << "No grades are recorded." << endl;
    }
    return 0;
    }
```





6. Conclusion

Learning how to control structures

7. Rubric